

Verified Steel Operator GmbH

Verification Readiness & Evidence Assurance Dossier

Prepared for Independent Accredited Verifier Review

OPERATOR READINESS STATUS

OPERATOR_PREPARATION_COMPLETE

DIAGNOSTIC SCORE

100 / 100

Report ID:	report_aaa
Case ID:	case_verifier_grade_fixture
Product Delivery Tier:	Premium Dossier Pack (V5)
Dossier Release Iteration:	Iteration #5 (Sealed Release)
Product Engine Version:	V5.0 (Definitive)
Generated At:	2027-01-15T12:00:00.000Z
Reporting Year & Period:	2026-ANNUAL
Operator Name:	Verified Steel Operator GmbH
Installation Name:	Verified Integrated Steel Installation

SECURE TRUST STAMP & KMS SIGNATURE RECORD

Case Snapshot SHA-256 Hash: 8dfbb38fc2291f651c27fabcc845abea676dcb448c2feb1fcb884afb1891a675

Calculation Root Hash: 7e12ba9592177cc10e0e4541bb103905e0345303fa794949881b86ec2ab48f61

Manifest SHA-256 Hash: NOT_AVAILABLE

Sealed Package Hash: NOT_AVAILABLE

KMS Key Version: NOT_AVAILABLE

KMS Signature Prefix: NOT_AVAILABLE

Sealed Package Integrity: All 25 controlled package components frozen & digitally signed.

This operator-prepared package supports preparation for independent CBAM review. It is not an independent verification opinion, a reasonable-assurance conclusion, a customs decision, an EU approval, a CBAM Registry submission, or a guarantee of acceptance.

CHAPTER I: EXECUTIVE & LEGAL OVERVIEW

Document Control & Legal Framework

2. Document Control

Control Parameter	
Report ID	report_aa
Case ID	case_verifier_grade_fixture
Release Version	V5
Generated Timestamp	2027-01-15T12:00:00.000Z
Ruleset Version	EU-CBAM-DEFINITIVE-2026
Calculation Engine Version	3.0.0
Verification Class	Prepared for Independent Accredited Review

3. Legal Status and Reliance Boundary

This operator-prepared package supports preparation for independent CBAM review. It is not an independent verification opinion, a reasonable-assurance conclusion, a customs decision, an EU approval, a CBAM Registry submission, or a guarantee of acceptance.

CBAMVALID INTERNAL AUTOMATED READINESS ASSESSMENT

This pack represents the operator's internal automated readiness work. Independent accredited verifier review status remains NOT_REVIEWED. ACCREDITED_VERIFICATION_OPINION is required for customs submission.

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CHAPTER II: DIAGNOSTIC AUDIT & READINESS

Executive Board & Scoring Matrix

5. Executive Decision Board

The following summary table outlines the key metrics and decisions for senior executive/CFO review before independent verifier handover.

Readiness Score						
100/100	0	0	0	0	0	READY_FOR_ACCREDITED_VERIFIER_ENGAGEMENT

RECOMMENDED ACTION CONTEXT

All hard gates passed with 100% weighted dimension coverage. Operator preparation is complete and ready for accredited verifier engagement.

Assessed coverage: 100% of weighted dimensions. Passed within assessed: 100%.

6. Readiness Score and Hard Gates

The diagnostic score is the absolute sum of weighted dimension scores out of 100. NOT_ASSESSED dimensions contribute zero and prevent OPERATOR_PREPARATION_COMPLETE. Hard blockers override the score and force NOT_READY.

Readiness Dimension				
Identity	10%	100%	10%	6 / 6
Scope And Methodology	15%	100%	15%	3 / 3
Activity Data	15%	100%	15%	5 / 5
Evidence	20%	100%	20%	18 / 18
Calculation Integrity	15%	100%	15%	3 / 3
Allocation And Reconciliation	10%	100%	10%	3 / 3
Data Quality And Uncertainty	10%	100%	10%	3 / 3
Package Integrity	5%	100%	5%	2 / 2

CHAPTER III: INSTALLATION & SYSTEM BOUNDARY

Facility Identity & CN Mapping

7. Operator and Installation Identity

Identity Attribute	
Operator Name	Verified Steel Operator GmbH
Importer Name	CBAMValid Importer B.V.
EORI Number	NL123456789AB
Installation Name	Verified Integrated Steel Installation
Country of Origin	DE

8. Reporting Period Assessment

Calendar-aware analysis of the reporting period for definitive annual verification eligibility. Period end must not exceed the assessment timestamp.

Reporting Attribute	
Reporting Year	2026
Reporting Period Type	DEFINITIVE_ANNUAL
Covered Dates	2026-01-01 to 2026-12-31
Covered Days count	365 days
Expected Days count	365 days
Completeness Percentage	100%
Completeness Status	PASSED
Assessment Timestamp	2027-01-15T12:00:00.000Z
Definitive Annual Eligible	YES (PASSED)

9. Goods and CN Classification

The case includes the following CN-coded goods classifications:

Good Index			
1	72011011	IRON_AND_STEEL	60 t
2	72011019	IRON_AND_STEEL	40 t

10. Installation and System Boundary

DECLARED SYSTEM BOUNDARY
Coke preparation, sinter plant, blast furnace, basic oxygen furnace, casting and finishing operations within the controlled installation boundary.

11. Production Processes and Functional Units

Controlled production processes and default boundaries of matching sectors:

IRON AND STEEL
Legal Status: IN_FORCE. Default boundaries: Coke oven, sinter or pellet preparation, blast furnace, basic oxygen furnace, direct reduction, electric arc furnace, casting and finishing processes within the installation boundary.

CHAPTER IV: MATERIAL ACTIVITY & EVIDENCE LINEAGE

Activity Ledger & Document Lineage

12. Material Input Register

Material activity data inputs required for the production route:

Input Path			
directEmissions	80	tCO2e	Primary
electricityConsumed	100	MWh	Primary
gridEmissionFactor	0.4	tCO2e/MWh	Primary

13. Evidence Sufficiency Matrix

Requirement-level analysis of evidence linkages. PARTIALLY_SUPPORTED or missing evidence blocks sealing.

Req ID					
REQ-OP-NAME	exporterIdentity.legalName	11111111-1111-41 11-8111-22222222 2222	Supported	365 / 365	Pass
REQ-OP-ADDR	exporterIdentity.address	11111111-1111-41 11-8111-22222222 2222	Supported	365 / 365	Pass
REQ-IM-NAME	importerIdentity.legalName	11111111-1111-41 11-8111-22222222 2222	Supported	365 / 365	Pass
REQ-IM-EORI	importerIdentity.eoriNumber	11111111-1111-41 11-8111-22222222 2222	Supported	365 / 365	Pass
REQ-INST-NAM E	installation.name	11111111-1111-41 11-8111-22222222 2222	Supported	365 / 365	Pass
REQ-INST-COU NTRY	installation.country	11111111-1111-41 11-8111-22222222 2222	Supported	365 / 365	Pass
REQ-INST-ROU TE	installation.productionRoute	11111111-1111-41 11-8111-22222222 2222	Supported	365 / 365	Pass
REQ-INST-BOU NDS	installation.systemBoundaries	—	Supported	1.0 / 1.0	Pass
REQ-PERIOD-Y EAR	reportingPeriod.year	11111111-1111-41 11-8111-22222222 2222	Supported	365 / 365	Pass
REQ-GOOD-CN -0	goods.0.cnCode	11111111-1111-41 11-8111-22222222 2222	Supported	365 / 365	Pass
REQ-GOOD-VO L-0	goods.0.productionVolume	11111111-1111-41 11-8111-44444444 4444	Supported	365 / 365	Pass

Req ID					
REQ-GOOD-AL-LOC-0	goods.0.allocationShare	11111111-1111-4111-8111-222222222222, 11111111-1111-4111-8111-444444444444	Supported	365 / 365	Pass
REQ-GOOD-CN-1	goods.1.cnCode	11111111-1111-4111-8111-222222222222	Supported	365 / 365	Pass
REQ-GOOD-VO-L-1	goods.1.productionVolume	11111111-1111-4111-8111-444444444444	Supported	365 / 365	Pass
REQ-GOOD-AL-LOC-1	goods.1.allocationShare	11111111-1111-4111-8111-222222222222, 11111111-1111-4111-8111-444444444444	Supported	365 / 365	Pass
REQ-DIR-EM	directEmissions	11111111-1111-4111-8111-555555555555	Supported	365 / 365	Pass
REQ-ELEC-CO-N	electricityConsumed	11111111-1111-4111-8111-333333333333	Supported	365 / 365	Pass
REQ-ELEC-FAC	gridEmissionFactor	11111111-1111-4111-8111-333333333333	Supported	365 / 365	Pass

14. Evidence Register

Evidence ID					
11111111-1111-4111-8111-222222222222	customs-declaration-classification.txt	Customs Declaration	Approved	Clean	131f55d589...
11111111-1111-4111-8111-333333333333	electricity-utility-invoice.txt	Utility Bill	Approved	Clean	7e9c3cdb25...
11111111-1111-4111-8111-444444444444	production-reconciliation-ledger.txt	Production Reconciliation Report	Approved	Clean	c5ecc502cc...
11111111-1111-4111-8111-555555555555	verified-monitoring-package.txt	Primary Monitoring And Customs Package	Approved	Clean	52f88d25d5...

15. Data Lineage Matrix

Cryptographic audit path linking physical evidence hash to final calculation results:

Evidence ID			
11111111-1111-4111-8 111-222222222222	131f55d58925...	importerIdentity.eoriNumber, goods.0.cnCode, goods.1.cnCode, exporterIdentity.legalName, exporterIdentity.address, importerIdentity.legalName, importerIdentity.eoriNumber, installation.name, installation.country, installation.productionRoute, reportingPeriod.year, reportingPeriod.quarter, goods.0.cnCode, goods.0.allocationShare, goods.1.cnCode, goods.1.allocationShare	None
11111111-1111-4111-8 111-333333333333	7e9c3cdb2568...	electricityConsumed, gridEmissionFactor	CBAM_INDIR
11111111-1111-4111-8 111-444444444444	c5ecc502ccc8...	goods.0.productionVolume, goods.0.allocationShare, goods.1.productionVolume, goods.1.allocationShare	CBAM_GOOD_, CBAM_GOOD_, CBAM_GOODS
11111111-1111-4111-8 111-555555555555	52f88d25d5eb...	directEmissions	CBAM_TOTAL

CHAPTER V: CALCULATION & ALLOCATION ANNEX

Direct, Indirect, Precursor & Per-Good Quantifications

16. Direct Emissions

Direct greenhouse gas emissions within the installation boundary as declared in the case data:

Emissions Category				
Installation Direct Scope	PRIMARY	80 tCO2e	Documented direct measurement and reconciled production ledger	80 tCO2e

17. Indirect Emissions

Indirect emissions associated with imported electricity consumed in production processes:

Indirect Emissions Component				
Electricity Consumed	PRIMARY	100 MWh	0.4 tCO2e/MWh	40 tCO2e

18. Precursors

NOT_APPLICABLE: No precursors declared for this production route.

19. Allocation and Per-good Results

Good				
1	72011011	0.6	72	1.2
2	72011019	0.4	48	1.2

20. Calculation Integrity and Reconciliation

Parameter			
Allocation Share Total	1	fraction	RECONCILED
Allocation Reconciliation Delta	0	fraction	ZERO_DELTA_PASS

CHAPTER VI: VERIFIER HANDOVER & TECHNICAL ANNEXES

Governance, Audit Trails & Manifest Sign-Off

21. Data Quality, Uncertainty, and Missing Data

The operator-supplied activity data and monitoring instruments are evaluated against definitive-period embedded-emissions methodology under Commission Implementing Regulation (EU) 2025/2547 of 10 December 2025 laying down rules for the application of Regulation (EU) 2023/956 as regards the methods for the calculation of emissions embedded in goods (CELEX 32025R2547). No data gaps were auto-filled using unverified figures.

Declared Data Stream			
Direct Emissions	Documented direct measurement and reconciled production ledger	11111111-1111-4111-8111-5555555555555	COMPLIANT
Electricity Consumed	Documented direct measurement and reconciled production ledger	11111111-1111-4111-8111-3333333333333	COMPLIANT
Grid Emission Factor	Documented direct measurement and reconciled production ledger	11111111-1111-4111-8111-3333333333333	COMPLIANT

22. Methodology Decision Register

Topic			
Precursor Scope	No qualifying precursor line is declared for the fixture goods and controlled route.	The controlled bill of materials and production route contain no separate precursor goods requiring an additional precursor line in this fixture.	Accepted
Goods Emissions Allocation	Allocate installation emissions using documented product mass shares of 0.6 and 0.4.	The two goods are produced in the same controlled period and the reconciled product ledger provides complete mass shares summing to one.	Accepted

23. Findings Register

No open findings or non-conformities are detected.

24. Corrective Action Plan

All corrective actions are closed or not required.

25. EU Verification Template Crosswalk

Req ID				
CW-OP-NAME	Article 8 & Annex VI.1(a)	Operator Name and Registration Address	OPERATOR	COMPLETE
CW-INST-INFO	Article 6 & Annex	Installation Name, address and UN/LOCODE coordinates	OPERATOR	COMPLETE
CW-REP-PERIOD	Article 8	Report ID and Reporting period	CBAMVALID_SYSTEM	COMPLETE
CW-VER-ID	Annex VI.2	Verifier identity, address, NAB country, accreditation scope and expiry	INDEPENDENT_VERIFIER	PENDING_EXTERNAL_VERIFIER
CW-VER-VISIT	Article 6(2)	Installation physical visit date or legal waiver justification basis	INDEPENDENT_VERIFIER	PENDING_EXTERNAL_VERIFIER

Req ID				
CW-GOOD-QTY-0	Annex VI.1(c)	Production quantity and CN-code verification for Good 1	OPERATOR	COMPLETE
CW-GOOD-QTY-1	Annex VI.1(c)	Production quantity and CN-code verification for Good 2	OPERATOR	COMPLETE
CW-DIR-EM	Annex VI.1(d)	Installation direct emissions monitoring plan data and methodology	OPERATOR	COMPLETE
CW-ALLOC-METHOD	Annex II	Emissions attribution methodology and allocation share factors	OPERATOR	COMPLETE
CW-NON-ASSOC	Annex II.3	Identification of non-associated goods/emissions/energy flows	OPERATOR	COMPLETE
CW-PRECURSOR-INST	Annex II.4.1	Precursor-producing installation identity, country and registration when actual emissions are used	OPERATOR	NOT_APPLICABLE
CW-VER-MISSTATEMENTS	Annex VI.3	Verifier audit trace of material misstatements detected and corrected during verification	INDEPENDENT_VERIFIER	PENDING_EXTERNAL_VERIFIER
CW-VER-NONCONFORMITIES	Annex VI.3	Verifier audit trace of non-conformities with monitoring rules detected and corrected	INDEPENDENT_VERIFIER	PENDING_EXTERNAL_VERIFIER
CW-VER-OPINION	Annex VI.4	Verifier final reasonable-assurance opinion statement and sign-off	INDEPENDENT_VERIFIER	PENDING_EXTERNAL_VERIFIER

26. Verifier Handover Checklist

The following mandatory verifier-completion items require manual audit and signature by an independent accredited verifier:

Verifier Reserved Field			
Accredited Verifier Name & ID	ISO 14065 / EU AVR Accreditation	VERIFIER_COMPLETION_REQ UIRED	Must be recorded post-audit
Accreditation Body & National Authority	EU National Accreditation Body (NAB)	VERIFIER_COMPLETION_REQ UIRED	Must be recorded post-audit
Site-Visit Execution & Date Range	Article 29 EU AVR 2018/2067	VERIFIER_COMPLETION_REQ UIRED	Physical or virtual site visit log
Materiality Threshold & Misstatement Finding	5% Materiality Threshold Assessment	VERIFIER_COMPLETION_REQ UIRED	Conformity decision statement
Verification Opinion & Assurance Class	Reasonable Assurance Class Statement	VERIFIER_COMPLETION_REQ UIRED	Opinion certificate required
Verification Report Certificate Ref	Official Accreditation Registry Number	VERIFIER_COMPLETION_REQ UIRED	Certificate reference code
Lead Verifier Digital Signature	Cryptographic / Qualified Digital Signature	VERIFIER_COMPLETION_REQ UIRED	Must sign final verification report
Independent Reviewer Sign-off	Dual-Control Independent Peer Review	VERIFIER_COMPLETION_REQ UIRED	Peer reviewer verification

27. Package Manifest and Digital Integrity

Integrity Parameter	
Case Snapshot SHA-256 Hash	8dfbb38fc2291f651c27fabcc845abea676dcb448c2feb1fcb884afb1891a675
Calculation Root Hash	7e12ba9592177cc10e0e4541bb103905e0345303fa794949881b86ec2ab48f61
Manifest SHA-256 Hash	NOT_AVAILABLE
Sealed Package SHA-256 Hash	NOT_AVAILABLE
KMS Key Version Reference	NOT_AVAILABLE
KMS Signature Algorithm	NOT_AVAILABLE
KMS Signature Base64	NOT_AVAILABLE
Schema Specification	CBAMVALID-DOSSIER-5.0
Digital Signature ID	report_aa
Cryptographic Security Class	FIPS 140-2 Level 3 KMS Sealed Hash
Public Verification State	ACTIVE

28. Version Comparison

This register tracks the history of released and sealed package versions under this case scope:

No previous sealed release exists.

Version				
V5	2027-01-15T12:00:00.000Z	Dossier correction/update release.	OPERATOR_ADMIN	ACTIVE_RELEASE

29. Sign-off and Limitations

The operator hereby signs off on this verifier-preparation dossier as complete and accurate to the best of their knowledge. This package does not constitute a legally binding accredited verifier certificate.

Sign-off Role			
Operator Author / Preparer	NOT_PROVIDED	OPERATOR_PREPARED	2027-01-15
Internal Environmental Reviewer	NOT_PROVIDED	REVIEW_REQUIRED	NOT_AVAILABLE
Independent Accredited Verifier	NOT_AVAILABLE	VERIFIER_COMPLETION_REQUIRED	NOT_AVAILABLE

30. Technical Annex Index

The sealed ZIP package contains the following 25 controlled components required for independent verifier review:

Filename in Sealed Package ZIP		
CBAMValid Verification Readiness & Evidence Assurance Dossier.pdf	PDF	Primary executive & verifier readiness dossier
Complete Dossier Compilation.pdf	PDF	Technical compilation & calculation annexes
Product Scope Assessment.pdf	PDF	System boundary & sectoral scope register

Filename in Sealed Package ZIP		
CN Code Reasoning.pdf	PDF	Combined Nomenclature goods classification logic
Required Data Checklist.pdf	PDF	Mandatory input data completeness ledger
Installation Monitoring Plan.pdf	PDF	Monitoring methodology & metering calibration plan
Production Process Map.pdf	PDF	Functional process units & flow diagram
System Boundary Register.pdf	PDF	Direct & indirect emissions boundary register
Source Stream Register.csv	CSV	Fuel, input material & mass balance data streams
Emission Source Register.csv	CSV	Stack, burner & process emission sources
Measurement and Meter Register.csv	CSV	Metering instruments & uncertainty log
Activity Data Ledger.csv	CSV	Daily/monthly raw activity data records
Evidence Register.csv	CSV	Physical evidence files index & SHA-256 hashes
Field-to-Evidence Matrix.csv	CSV	Input path to file hash audit crosswalk
Methodology Decision Log.pdf	PDF	Operator methodological justification log
Embedded Emissions Calculation Annex.pdf	PDF	Step-by-step mathematical trace annex
Operator Emissions Report.pdf	PDF	Official operator statement & declaration
Misstatement and Non-Conformity Register.csv	CSV	Quality controls & findings register
Corrective Action Log.csv	CSV	Remediation action tracking ledger
O3CI Field Mapping.csv	CSV	Registry export data field crosswalk
Calculation Trace.json	JSON	Machine-readable cryptographic node hash tree
Verifier Workspace.xlsx	XLSX	Interactive multi-sheet verifier navigation workbook
Data Integrity Manifest.json	JSON	Cryptographic package manifest & hash index
Manifest Signature.sig	SIG	KMS detached signature over the integrity manifest
Supporting_Evidence/	DIR	Tenant-bound source evidence binaries